# Example



Subject to

$x \geq 0$

$y \leq 0$

$x + 3y \leq 15$

$2x + 3y \geq 100$

$3x + 4y \geq 80$

$\rightarrow$

$-x \leq 0$

$-y \leq 0$

$-x - 3y \leq -15$

$2x + 3y \geq 100$

$3x + 4y \geq 80$

**x=0, y=5**

$x + 3y - 15 \rightarrow \leq 15$

$2x + 3y - 25 \rightarrow \geq 100$

$3x + 4y - 20 \rightarrow \geq 80$

$x \leq 0$

$y \leq 0$

*Minimum value:*